

Test Pilot IQ (dTQ) — Demo Guide

Version: 1.0.0 **Last Updated:** 2026-02-04 **Status:** Production Demo **Part of:** Digital Workplace AI Product Suite

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1. Executive Overview

What is Test Pilot IQ?

Test Pilot IQ (dTQ) is an **AI-powered QA & testing intelligence platform** that gives engineering teams unified, real-time visibility into test health, feature coverage, and quality trends. It is part of the Digital Workplace AI product suite.

The Problem It Solves

QA teams today lack a single pane of glass that connects test execution data, feature coverage, risk analysis, and quality trends — and presents them in a way that is meaningful to every stakeholder, from executives to individual engineers. Test Pilot IQ solves this by providing:

- **Role-based dashboards** — the same dataset, presented through the lens of C-Suite executives, QA Managers, and Tech Leads.
- **Real-time simulation** — continuous test run generation that mimics a live CI/CD pipeline.
- **AI-powered insights** — a floating chat assistant backed by Claude AI and RAG (Retrieval-Augmented Generation) that can answer natural-language questions about test data.
- **Actionable navigation** — every metric, chart point, feature, and category is clickable, opening detailed drill-down modals.

Key Value Propositions

Value	Description
3 Persona Views	C-Suite sees ROI and risk reduction; QA Manager sees team pass rates and regression times; Tech Lead sees flaky test rates and pipeline efficiency.

Real-Time Simulation	A new test run is generated every 8–15 seconds, with metrics fluctuating to mimic a live environment.
AI Chat with RAG	Ask natural-language questions. The system embeds your query, performs semantic search across 130 knowledge base items, and generates a contextual response via Claude.
Actionable Navigation	Click any metric card, chart data point, feature row, or category header to open a detailed drill-down modal. Chat link cards navigate you directly to the relevant page and modal.

Tech Stack

Technology	Version	Purpose
Next.js	16.x	React framework with App Router
TypeScript	5.x	Type safety
Supabase	@supabase/supabase-js	Database + pgvector
Claude AI (Anthropic)	Sonnet 4	Chat responses with RAG context
OpenAI	text-embedding-3-small	1536-dim embeddings for semantic search
Tailwind CSS	4.x	Styling
Framer Motion	12.x	UI animations
Recharts	3.x	Charts and visualizations
GSAP	3.x	Advanced animations

2. Access & Navigation

URLs

Environment	Dashboard	History	
Local Dev	http://localhost:3004/dtq/dashboard	http://localhost:3004/dtq/history	http://localhost:3004/dtq/chat
Production	<a href="https://<domain>/dtq/dashboard">https://<domain>/dtq/dashboard	<a href="https://<domain>/dtq/history">https://<domain>/dtq/history	<a href="https://<domain>/dtq/chat">https://<domain>/dtq/chat

Note: The app uses `basePath: "/dtq"` in `next.config.ts`. Port 3004 is the default dev port.

Sidebar Navigation

The left sidebar provides navigation to the three main pages:

Icon	Label	Route	Description
LayoutDashboard	Dashboard	<code>/dtq/dashboard</code>	Primary metrics, KPIs, trend charts, feature coverage

Clock	Metrics History	/dtq/history	30-day trend charts and summary averages
FileText	Test Reports	/dtq/reports	Test run table with filtering, sorting, pagination

Persona Switcher

Located in the sidebar, below the navigation links. Three persona buttons are displayed:

Persona	Icon	ID
C-Suite	Crown	csuite
QA Manager	Users	manager
Tech Lead	Code	techlead

Behavior: Clicking a persona button updates the entire dashboard — all features, KPIs, test runs, and daily metrics change to reflect the selected persona's data. The active persona is highlighted with the accent color.

Live / Pause Toggle

Located in the top-right corner of every page. Displays a pulsing green dot when live.

- **Live (default):** New test runs are generated every 8–15 seconds. Metrics fluctuate with small variance.
- **Paused:** Simulation stops. Data remains static until resumed.

3. Personas — Full Spectrum

Test Pilot IQ presents the same underlying platform through three distinct lenses. Each persona has its own set of features, categories, KPIs, and test runs.

3a. C-Suite (Executive)

Role Description: The C-Suite persona is designed for VPs of Engineering, CTOs, and other executive stakeholders who care about business impact — revenue protection, customer experience, compliance posture, and strategic ROI on test automation.

Summary: 16 features | 5 categories | 7 KPIs

Categories & Features

Revenue Platform (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
cs-f1	Payment Gateway	96%	Fully Automated	12	98%	95
cs-f2	Subscription Billing	93%	Fully Automated	18	96%	92
cs-f3	Revenue Analytics	91%	Fully Automated	15	97%	90

cs-f4	Pricing Engine	88%	Partially Automated	22	94%	88
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Customer Experience (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
cs-f5	Customer Portal	94%	Fully Automated	14	97%	93
cs-f6	NPS Tracking	89%	Partially Automated	20	93%	86
cs-f7	Support Escalation	92%	Fully Automated	16	96%	89
cs-f8	Onboarding Journey	90%	Fully Automated	19	95%	91

Market Expansion (3 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
cs-f9	Multi-Region Deployment	87%	Partially Automated	24	92%	94
cs-f10	Localization Engine	85%	Partially Automated	28	90%	82
cs-f11	Partner Integration	91%	Fully Automated	17	95%	87

Strategic Compliance (3 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
cs-f12	SOC 2 Validation	97%	Fully Automated	10	99%	96
cs-f13	GDPR Data Handling	95%	Fully Automated	13	98%	94
cs-f14	Audit Trail System	93%	Fully Automated	16	96%	91

Digital Transformation (2 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
cs-f15	AI Feature Pipeline	82%	Partially Automated	30	89%	97
cs-f16	Legacy Migration	84%	Partially Automated	26	91%	85

KPIs (7 metrics)

Key	Label	Value	Unit	Trend	Trend Value
releaseVelocity	Release Velocity	+35	%	up	+8%
timeToMarket	Mean Time to Market	12	days	down	-4 days
automationROI	Automation ROI	285	%	up	+45%

escapeRate	Defect Escape Rate	2.1	%	down	-0.8%
incidentsPrevented	Incidents Prevented	47	count	up	+12
riskReduction	Risk Reduction	68	%	up	+15%
capacityUnlocked	Capacity Unlocked	42	%	up	+8%

Use Case Scenario

"As a VP of Engineering, I want to see the ROI on test automation investment. I open the Dashboard with the C-Suite persona selected. The Automation ROI metric card shows 285% — I click it and the MetricDrillDownModal reveals 30-day trends, period comparison, and category breakdown. I then ask the AI Chat 'What is our defect escape rate trend?' and receive a contextual answer with a link card that navigates me to the Metrics History page."

3b. QA Manager

Role Description: The QA Manager persona focuses on team performance and operational quality — pass rates, regression execution times, test case reuse, escalation rates, and environment stability.

Summary: 46 features | 10 categories | 9 KPIs

Categories & Features

Advanced / Enterprise (5 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f1	xAPI / LRS Integration	92%	Fully Automated	45	94%	85
f2	Multi-Tenant Configuration	88%	Partially Automated	32	91%	92
f3	Advanced Compliance Tracking	95%	Fully Automated	18	97%	88
f4	Custom Branding Engine	78%	Partially Automated	25	89%	65
f5	Enterprise SSO	96%	Fully Automated	12	98%	95

Learning Experience (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f6	Adaptive Learning Paths	85%	Partially Automated	38	88%	90
f7	Personalized Recommendations	91%	Fully Automated	22	93%	82

f8	Skill Gap Analysis	87%	Partially Automated	28	90%	78
f9	Learning Goals Tracking	93%	Fully Automated	15	96%	75

Reporting & Analytics (5 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f10	Executive Dashboard	94%	Fully Automated	20	95%	92
f11	Custom Report Builder	82%	Partially Automated	35	87%	85
f12	Learning Analytics	89%	Partially Automated	42	91%	88
f13	Compliance Reports	97%	Fully Automated	10	99%	94
f14	Export Center	90%	Fully Automated	16	94%	70

Course Catalog & Enrollment (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f15	Course Discovery	86%	Partially Automated	30	89%	85
f16	Self-Enrollment	95%	Fully Automated	14	97%	90
f17	Manager-Assigned Learning	88%	Partially Automated	24	92%	82
f18	Waitlist Management	79%	Partially Automated	20	86%	60

Social & Collaboration (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f19	Discussion Forums	83%	Partially Automated	26	88%	75
f20	Peer Reviews	91%	Fully Automated	18	93%	72
f21	Learning Groups	87%	Partially Automated	22	90%	68
f22	Social Sharing	76%	Partially Automated	15	85%	55

Course Creation & Management (5 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f23	SCORM Import	94%	Fully Automated	19	95%	88
f24	Content Authoring	81%	Partially Automated	34	86%	92

f25	Assessment Builder	89%	Partially Automated	28	91%	85
f26	Media Library	85%	Partially Automated	21	89%	78
f27	Course Versioning	92%	Fully Automated	16	94%	80

Admin & Configuration (6 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f28	User Management	96%	Fully Automated	12	98%	95
f29	Role-Based Access Control	93%	Fully Automated	18	96%	94
f30	Notification Settings	84%	Partially Automated	22	88%	65
f31	Integration Hub	78%	Partially Automated	36	85%	88
f32	Audit Logs	97%	Fully Automated	10	99%	90
f33	System Configuration	88%	Partially Automated	24	91%	85

Assessments & Certification (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f34	Quiz Engine	92%	Fully Automated	20	94%	88
f35	Certification Paths	90%	Fully Automated	16	93%	92
f36	Proctoring Support	75%	Partially Automated	32	84%	78
f37	Certificate Designer	86%	Partially Automated	18	90%	70

Events & Live Sessions (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
f38	Webinar Integration	82%	Partially Automated	28	87%	82
f39	ILT Scheduling	89%	Partially Automated	22	92%	85
f40	Attendance Tracking	94%	Fully Automated	14	96%	78
f41	Virtual Classroom	77%	Partially Automated	38	85%	88

Core User Journeys (5 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
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f42	User Onboarding	91%	Fully Automated	24	93%	95
f43	Course Completion Flow	95%	Fully Automated	12	97%	98
f44	Profile Management	88%	Partially Automated	18	91%	72
f45	Mobile Experience	79%	Partially Automated	42	86%	90
f46	Search & Discovery	84%	Partially Automated	30	88%	85

KPIs (9 metrics)

Key	Label	Value	Unit	Trend	Trend Value
passRate	Test Pass Rate	94.2	%	up	+2.3%
autoManualRatio	Automated vs Manual	8.5:1	ratio	up	+0.5
effectiveness	Test Case Effectiveness	0.87	score	stable	0.0
regressionTime	Regression Execution	42	min	down	-8 min
firstRunPass	First-Run Pass Rate	87.5	%	up	+4.2%
escalationRate	Escalation Rate	12.3	%	down	-2.1%
testReuse	Test Case Reuse	76.8	%	up	+5.3%
blockerDefects	Blocker Defects	3	count	down	-2
envUptime	Environment Uptime	99.2	%	stable	0.0%

Use Case Scenario

"As a QA Manager, I want to monitor my team's pass rates and identify flaky test areas. I open the Dashboard and see the Test Pass Rate KPI at 94.2% with an upward trend. I click the Regression Execution metric card and see it's down to 42 minutes — an 8-minute improvement. In the Feature Coverage section, I expand 'Core User Journeys' and click 'Mobile Experience' (risk score 42) to see its 14-day history chart and open defects."

3c. Tech Lead (Engineer)

Role Description: The Tech Lead persona provides engineering depth — CI/CD pipeline health, API infrastructure reliability, performance engineering metrics, security testing posture, and DevOps automation coverage.

Summary: 18 features | 5 categories | 8 KPIs

Categories & Features

CI/CD Pipeline (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
tl-f1	Build Orchestration	88%	Fully Automated	25	91%	90
tl-f2	Deploy Pipeline	85%	Partially Automated	32	88%	92
tl-f3	Rollback Automation	82%	Partially Automated	35	86%	88
tl-f4	Feature Flags	90%	Fully Automated	20	93%	78

API Infrastructure (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
tl-f5	REST API Gateway	92%	Fully Automated	18	94%	95
tl-f6	GraphQL Layer	79%	Partially Automated	38	85%	88
tl-f7	Rate Limiting	86%	Fully Automated	22	92%	82
tl-f8	API Versioning	83%	Partially Automated	28	89%	76

Performance Engineering (3 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
tl-f9	Load Testing Framework	78%	Partially Automated	40	84%	90
tl-f10	Memory Profiler	75%	Partially Automated	36	82%	85
tl-f11	CDN Optimization	91%	Fully Automated	15	95%	80

Security Testing (3 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
tl-f12	Penetration Test Suite	70%	Partially Automated	45	80%	96
tl-f13	Dependency Scanner	93%	Fully Automated	16	94%	88
tl-f14	Auth Token Validation	89%	Fully Automated	20	92%	94

DevOps Automation (4 features)

ID	Feature	Coverage	Status	Risk	Pass Rate	Impact
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tl-f15	Container Orchestration	84%	Partially Automated	30	88%	92
tl-f16	Log Aggregation	87%	Fully Automated	19	91%	78
tl-f17	Infra-as-Code	81%	Partially Automated	33	86%	90
tl-f18	Secrets Management	96%	Fully Automated	10	98%	95

KPIs (8 metrics)

Key	Label	Value	Unit	Trend	Trend Value
flakyRate	Flaky Test Rate	3.2	%	down	-1.5%
avgExecution	Avg Execution Time	4.2	min	down	-0.8 min
tokenCost	Token Usage Cost	0.42	\$	down	-\$0.15
contextHitRate	Context Hit Rate	94.5	%	up	+2.3%
toolCallSuccess	Tool Call Success	97.8	%	up	+1.2%
parallelEfficiency	Parallel Efficiency	78.5	%	up	+5%
automationCoverage	Automation Coverage	93.3	%	up	+2.1%
totalTests	Total Test Cases	1247	count	up	+86

Use Case Scenario

"As a Tech Lead, I want to see flaky test rates and pipeline efficiency. I switch to the Tech Lead persona and immediately see the Flaky Test Rate at 3.2% (trending down). I click the 'CI/CD Pipeline' category in Feature Coverage and the CategoryAnalyticsModal shows a bar chart comparing Build Orchestration, Deploy Pipeline, Rollback Automation, and Feature Flags. I notice Rollback Automation has a risk score of 35, so I click it to see its 14-day trend and 2 open defects."

4. Dashboard Page — Full Walkthrough

The Dashboard (/dtq/dashboard) is the primary page. It is divided into six sections from top to bottom.

4.1 Header

- **Title:** "Test IQ Dashboard" (with "Dashboard" in accent pink #ff3366)
- **Subtitle:** "AI-Powered Testing Intelligence for Enterprise Quality"
- **Live Indicator:** Top-right, shows pulsing green dot when live, last update timestamp, and toggle button.

4.2 Persona Card

Displays the currently selected persona with its icon, name, title, and description. Updates when the persona is switched via the sidebar.

4.3 Primary Metrics Grid (4 cards)

Four `MetricCard` components in a responsive grid (1 column on mobile, 2 on tablet, 4 on desktop):

Card	Value (Manager Example)	Subtitle	Icon	Click Action
Total Features	46	"88.5% avg coverage"	Layers	Opens <code>MetricDrillDownModal</code>
Automation Rate	43%	"20 fully automated"	Cpu	Opens <code>MetricDrillDownModal</code>
Risk Distribution	"3 High"	"25 Med / 18 Low"	AlertTriangle	Opens <code>MetricDrillDownModal</code>
Open Defects	35	"Across all features"	Bug	Opens <code>MetricDrillDownModal</code>

Note: Values shown are for the QA Manager persona. C-Suite and Tech Lead personas display different computed values.

4.4 High Risk Banner

Appears when any features have a risk score ≥ 40 . Displays the top 3 highest-risk features as clickable cards.

- Each card shows: feature name, risk score, coverage %, and pass rate.
- Clicking a feature card opens the `FeatureDetailModal`.
- The banner can be expanded or dismissed.

Manager persona example (risk ≥ 40):

Feature	Risk Score	Coverage	Pass Rate
xAPI / LRS Integration	45	92%	94%
Learning Analytics	42	89%	91%
Mobile Experience	42	79%	86%

4.5 Persona-Specific KPI Grid

Displays role-specific metrics in a responsive grid (1/2/3 columns). Each KPI card is clickable and opens the `MetricDrillDownModal`.

- **C-Suite:** 7 KPIs (see Section 3a)
- **QA Manager:** 9 KPIs (see Section 3b)
- **Tech Lead:** 8 KPIs (see Section 3c)

Each card shows: label, value with unit, description, trend direction (up/down/stable), and trend value.

4.6 Trend Charts (7-Day)

Two charts side by side (stacked on mobile):

Chart	Type	Color	Data Source
Test Pass Rate Trend (7 Days)	Line chart	var(--accent-primary)	Last 7 entries of dailyMetrics.passRate
Automation Coverage (7 Days)	Area chart	var(--chart-secondary)	Last 7 entries of dailyMetrics.automationCoverage

Click interaction: Clicking any data point on either chart opens the **ChartDrillDownModal** with that point's value, date, trend analysis, and context comparison.

4.7 Feature Coverage Section

An expandable accordion of categories. Each category header shows:

- Category name
- Number of features
- Average coverage %
- High-risk feature count

Interactions:

- **Click category header:** Expands to show feature list; also serves as button to open **CategoryAnalyticsModal**.
- **Click feature row:** Opens the **FeatureDetailModal** for that feature.
- **Search:** A search input filters features by name across all categories.

5. Test Reports Page — Full Walkthrough

The Reports page (/dtq/reports) provides a comprehensive test execution history with filtering, sorting, pagination, and export capabilities.

5.1 Header

- **Title:** "Test Reports" (with "Reports" in accent pink)
- **Subtitle:** "Test execution history and issue tracking"
- **Live Indicator:** Top-right with live/pause toggle.
- **Refresh Button:** Manual refresh icon button next to the live indicator.

5.2 Summary Row (4 metric cards)

Card	Value	Icon	Interactive
Total Runs	Dynamic count	TestTube2	No
Pass Rate	Computed %	Activity	No
Failed Runs	Count of failed	XCircle	No
Avg Duration	Computed average (seconds)	Clock	No

These cards are informational only (not clickable).

5.3 Filter Bar

Three sections arranged horizontally:

Status Filter Tabs:

- **All** — Shows all test runs with count
- **Passed** — Shows only passed runs with count
- **Failed** — Shows only failed runs with count

Active tab is highlighted with an elevated background.

Search Input:

- Placeholder: "Search feature name..."
- Filters test runs by feature name (case-insensitive substring match)
- Has a search icon on the left

Export Buttons:

- **Export CSV** — Downloads a `.csv` file of all test runs
- **Export PDF** — Opens browser print dialog for PDF generation

5.4 Sortable Table

6 columns with sort functionality:

Column	Sortable	Sort Key	Description
Status	Yes	status	Pass/fail icon (CheckCircle2 green / XCircle red)
Feature Name	Yes	featureName	Feature name text
Tests	No	—	Passed/Total format (e.g., "18/18")
Duration	Yes	duration	Duration in seconds (hidden on mobile)
Executed	Yes	executedAt	Relative time (e.g., "2 days ago")
Issues	Yes	issues	Issue count badge or dash (hidden on mobile)

Sort behavior (3-click cycle):

1. First click: Sort ascending (↑ arrow shown)
2. Second click: Sort descending (↓ arrow shown)
3. Third click: Clear sort (returns to default order)

Unsorted columns show a faint ↑ arrow on hover.

Row click: Opens the **TestRunDetailModal** for that test run.

5.5 Row Highlight

When a new test run is generated by the real-time simulation, the newest row receives a brief highlight animation (`animate-highlight-row`) for 1.5 seconds.

5.6 Pagination

Displayed below the table when there are results:

- **Row indicator:** "1–10 of 40 runs"

- **Navigation buttons:** First (⏮), Previous (◀), Page numbers, Next (▶), Last (⏭)
- **Page size:** 10 rows per page (constant `ROWS_PER_PAGE = 10`)
- **Smart ellipsis:** Shows first, last, current, and neighbor page numbers with "..." for gaps
- Page resets to 1 when filters or sort change

5.7 Empty State

When no test runs match the current filters, a centered message appears:

- Search icon
- "No test runs match your filters"
- "Try adjusting the status filter or search query"

5.8 Cross-Modal Navigation

From the `TestRunDetailModal`, clicking "View Feature" navigates to the `FeatureDetailModal` for the associated feature. The test run modal closes and the feature modal opens.

6. Metrics History Page — Full Walkthrough

The Metrics History page (`/dtq/history`) provides 30-day trend analysis with four summary cards and four trend charts.

6.1 Header

- **Title:** "Team Performance Metrics" (with "Metrics" in accent pink)
- **Subtitle:** "Monitor team effectiveness and quality trends over the past 30 days"
- **Live Indicator:** Top-right with live/pause toggle.

6.2 Summary Cards (30-Day Averages)

Four large `MetricCard` components, each clickable to open `MetricDrillDownModal`:

Card	Icon	Metric Key	Description
Average Pass Rate	CheckCircle2	passRate	30-day average of <code>dailyMetrics.passRate</code>
First Pass Rate	Zap	firstRunPassRate	30-day average of <code>dailyMetrics.firstRunPassRate</code>
Defect Detection	Bug	defectDetection	30-day average of <code>dailyMetrics.defectDetection</code>
Test Effectiveness	Target	effectiveness	30-day average of <code>dailyMetrics.effectiveness</code>

6.3 Trend Charts (30-Day, Full Range)

Four charts in a 2x2 grid:

Chart	Type	Color	Data Source
Test Pass Rate Trend	Line	<code>var(--status-success)</code>	All 30 days of <code>passRate</code>

First Run Pass Rate	Line	var(--accent-primary)	All 30 days of firstRunPassRate
Defect Detection Percentage	Area	var(--risk-medium)	All 30 days of defectDetection
Test Case Effectiveness	Area	var(--chart-secondary)	All 30 days of effectiveness

Click interaction: Clicking any data point opens the **ChartDrillDownModal** with detailed analysis of that day's value, including comparison to previous/next days and percentile ranking.

7. AI Chat Assistant — Full Walkthrough

The AI Chat Assistant is a floating widget accessible from every page. It provides natural-language Q&A powered by Claude AI with RAG (Retrieval-Augmented Generation).

7.1 How to Open

A **floating action button** (FAB) is fixed at the bottom-right corner of the screen:

- Pink circle (`var(--accent-primary)`) with a `MessageCircle` icon
- Pulsing shadow effect
- Click to open the chat panel

When closed, an **unread badge** appears on the FAB if there are new messages (shows count, "9+" for 10+).

7.2 Chat Panel

When open, a 400px-wide (full-width on mobile) panel slides up from the bottom-right:

Header:

- AI sparkle icon with "AI Insights" title
- "Powered by intelligent analysis" subtitle
- Persona badge showing current persona (C-Suite / Manager / Tech Lead)
- Reset button (appears when messages exist) — clears all messages
- Close button (X)

Welcome State (no messages):

- Bot icon
- "Ask me about your testing metrics"
- "I have access to all 46 features with real-time data"

7.3 Quick Action Pills

Four pre-built query buttons, always visible above the input area:

ID	Icon	Label
high-risk	Target	High-risk features
feature-status	BarChart3	Feature status
automation-gaps	Settings	Automation gaps
quality-summary	TrendingUp	Quality summary

Clicking a pill sends the label as a user message and triggers the AI response pipeline.

7.4 Custom Questions

Type a question in the input field and press Enter or click the Send button. The input field auto-focuses when the panel opens.

7.5 RAG Pipeline

The full pipeline for each question:

1. **User message** is sent to `/api/dtq/chat` along with the current persona and conversation history.
2. **Embedding**: The message is converted to a 1536-dimensional vector using OpenAI's `text-embedding-3-small` model.
3. **Semantic search**: The vector is used to query `public.search_dtq_knowledge_semantic`, which performs cosine similarity search across 130 knowledge base items.
4. **Context injection**: Top matching items are injected as context into the Claude prompt.
5. **Claude response**: Claude Sonnet 4 generates a response using the RAG context, persona awareness, and conversation history.
6. **Link resolution**: The response text is processed by `link-resolver.ts` using three strategies:
 - o **Bold text extraction**: Matches `**Feature Name**` patterns against features and categories
 - o **Source-based mapping**: Maps RAG source types (`feature` , `category_summary` , `persona_kpi`) to navigation links
 - o **Keyword detection**: Detects "test run", "trend", "history" keywords to create report/history links
7. **Response rendered** with markdown formatting, source count, and related link cards.

7.6 Related Link Cards

Below AI responses, link cards may appear under a "Related" header. Each card shows:

- An icon matching the link target type (Target for feature, Boxes for category, Activity for metric, FileText for report, Clock for history, TestTube2 for test-run)
- Label (feature/category/metric name)
- Description (optional context)
- Chevron arrow on hover

Click behavior: Clicking a link card triggers `NavigationContext.dispatch()`, which:

1. Routes to the target page (`/dtq/dashboard` , `/dtq/reports` , or `/dtq/history`)
2. Sets a `pendingAction` with the link details
3. The target page's `useEffect` detects the pending action and opens the corresponding modal
4. The action is cleared after processing

7.7 Demo Mode vs RAG Mode

- **RAG Mode (default)**: When API keys are configured, the full RAG pipeline is used.
- **Demo Mode (fallback)**: When the API call fails, pre-built responses from `aiResponses` in `data.ts` are used. These provide rich markdown responses for the four quick actions.

7.8 Cross-Page Chat Persistence

The chat widget uses `ChatContext` to persist messages across page navigation. Messages are maintained in React state and survive route changes within the SPA.

7.9 Example Conversations by Persona

C-Suite:

- "What's our automation ROI?" → Response with 285% ROI, cost savings breakdown, link to Automation ROI metric
- "Show high-risk features" → Response listing AI Feature Pipeline (risk 30), Localization Engine (risk 28), Legacy Migration (risk 26)

QA Manager:

- "What's our regression execution time?" → Response with 42 min, -8 min improvement, link to regression metric
- "Which features have the most open defects?" → Response listing top defect features with links

Tech Lead:

- "What's the flaky test rate?" → Response with 3.2%, trending down, link to Flaky Test Rate metric
 - "Show CI/CD pipeline status" → Response with Build Orchestration, Deploy Pipeline details, links to features
-

8. Interactive Modals — Reference

All modals use a shared `BaseModal` component with animated entry/exit (Framer Motion), escape key handling, tab focus trapping, and customizable sizes (sm/md/lg/xl/full).

8.1 MetricDrillDownModal

Triggered by: Clicking any metric card (Dashboard KPIs, Dashboard primary metrics, History summary cards)

Contents:

- **Header:** Metric label, current value with unit, trend badge
- **30-Day Trend Chart:** Area chart showing the metric's daily values over 30 days
- **Statistics Row:** Current value, 30-Day Average, Peak, Low
- **Period Comparison:** Current value vs previous period value with percentage change
- **Category Breakdown:** Bar chart showing the metric broken down by feature category
- **Related Features:** List of features most impacted by this metric

8.2 ChartDrillDownModal

Triggered by: Clicking a data point on any trend chart (Dashboard or History)

Contents:

- **Header:** Metric label and date
- **Large Value Display:** The selected data point's value prominently displayed
- **Trend vs Previous Day:** Percentage change from the previous day
- **Percentile Ranking:** Where this value falls in the 30-day distribution
- **Context Comparison:** Previous day and next day values side by side
- **Contributing Factors:** List of factors that may have influenced this data point

8.3 FeatureDetailModal

Triggered by: Clicking a feature in Feature Coverage, High Risk Banner, or via cross-modal navigation

Contents:

- **Header:** Feature name, category, automation status badge
- **4 KPI Cards:** Coverage %, Risk Score, Open Defects (open/closed), Pass Rate
- **14-Day History Chart:** Line chart showing the feature's pass rate over the last 14 days
- **Recent Test Runs:** List of up to 10 most recent test runs for this feature, with status icons, test counts, duration, and timestamps
- **Open Defects Section:** List of current open defects with severity badges

8.4 CategoryAnalyticsModal

Triggered by: Clicking a category header in Feature Coverage

Contents:

- **Header:** Category name with feature count
- **4 Summary Cards:** Average coverage, average pass rate, high-risk feature count, total open defects
- **Feature Comparison Bar Chart:** Horizontal bar chart comparing coverage across all features in the category
- **Risk Distribution Pie Chart:** Pie chart showing the distribution of risk scores (high/medium/low)
- **Automation Breakdown:** Visual showing fully automated vs partially automated feature counts
- **Clickable Feature List:** Each feature row is clickable, opening the FeatureDetailModal (cross-modal navigation: category modal closes, feature modal opens)

8.5 TestRunDetailModal

Triggered by: Clicking a row in the Reports table

Contents:

- **Header:** Feature name, with status and date as description
 - **Status Badge:** Large pass/fail indicator
 - **4 Stat Cards:** Total Tests, Passed Tests, Failed Tests, Pass Rate (%)
 - **Issues Section (failed runs only):**
 - Severity filter tabs (All / High / Medium / Low)
 - Each issue shows: test case name, severity badge, error message
 - Expandable stack trace (click to toggle)
 - **Copy button:** Copies stack trace to clipboard
 - **View Feature Button:** Cross-modal navigation to the FeatureDetailModal for the associated feature
-

9. Real-Time Simulation

How It Works

The `useRealTimeSimulation` hook manages two simulation loops:

1. **Metric Fluctuation (every 10 seconds):** The latest day's daily metrics receive small random variance (± 0.5 – 1.5%) to simulate live metric movement.
2. **New Test Run Generation (every 8–15 seconds):** A new test run is generated with:
 - A random feature from the current persona's feature set
 - ~85% pass rate probability
 - 10–34 total tests per run

- 1–5 failed tests (if failed)
- 4–13 second duration
- Error messages and stack traces selected from pre-defined pools

Live / Pause Toggle

- **Location:** Top-right corner of every page
- **Live state:** Green pulsing dot, simulation active
- **Paused state:** Grey dot, simulation paused
- **Timestamp:** Shows "Updated X seconds ago" based on last update

Feature Pass Rate Updates

When a new test run is generated, the associated feature's pass rate is updated using an **exponential moving average** to provide smooth, realistic fluctuation.

Manual Refresh (Reports Page)

The Reports page includes a refresh button (rotating arrow icon) next to the Live Indicator. Clicking it triggers an immediate data refresh without waiting for the next simulation tick.

10. Data Export

CSV Export

Three export functions are available:

Test Runs CSV (`exportTestRunsToCSV`):

- File name: `test-reports-YYYY-MM-DD.csv`
- Columns: ID, Feature, Status, Total Tests, Passed, Failed, Duration (s), Executed At
- Includes all test runs in the current dataset

Features CSV (`exportFeaturesToCSV`):

- File name: `features-YYYY-MM-DD.csv`
- Columns: ID, Name, Category, Coverage (%), Status, Open Defects, Closed Defects, Risk Score, Pass Rate (%), Impact Score

Metrics CSV (`exportMetricsToCSV`):

- File name: `metrics-YYYY-MM-DD.csv`
- Columns: Date, Pass Rate (%), First Run Pass Rate (%), Defect Detection (%), Effectiveness (%), Automation Coverage (%)

PDF Export

Method: `exportTestRunsToPDF` generates an HTML document with styled tables and opens the browser's print dialog (`window.print()`). The user can then save as PDF from the print dialog.

Includes: Test run summary table with pass/fail status, test counts, duration, and execution timestamps.

File Naming Convention

All exported files follow the pattern: `{type}-YYYY-MM-DD.{ext}`

Examples:

- test-reports-2026-02-04.csv
 - features-2026-02-04.csv
 - metrics-2026-02-04.csv
-

11. TDD Test Scenarios (Full Spectrum)

The following test cases are structured as executable specifications using Given/When/Then format. They cover every interactive element in the application.

11.1 Navigation Tests

NAV-001: Sidebar navigation to Dashboard

```
Given the user is on any page
When they click "Dashboard" in the sidebar
Then the URL changes to /dtq/dashboard
And the Dashboard page content is displayed
And the "Dashboard" sidebar link is highlighted
```

NAV-002: Sidebar navigation to Metrics History

```
Given the user is on any page
When they click "Metrics History" in the sidebar
Then the URL changes to /dtq/history
And the History page content is displayed
And the "Metrics History" sidebar link is highlighted
```

NAV-003: Sidebar navigation to Test Reports

```
Given the user is on any page
When they click "Test Reports" in the sidebar
Then the URL changes to /dtq/reports
And the Reports page content is displayed
And the "Test Reports" sidebar link is highlighted
```

NAV-004: Persona switching to C-Suite

```
Given the user is on the Dashboard page
When they click the "C-Suite" persona button in the sidebar
Then all metric cards update to show C-Suite data (16 features, 7 KPIs)
And the persona card displays "C-Suite / Executive"
And the Feature Coverage section shows 5 categories
```

NAV-005: Persona switching to QA Manager

```
Given the user is on the Dashboard page
When they click the "QA Manager" persona button in the sidebar
Then all metric cards update to show Manager data (46 features, 9 KPIs)
```

And the persona card displays "QA Manager / Manager"
And the Feature Coverage section shows 10 categories

NAV-006: Persona switching to Tech Lead

Given the user is on the Dashboard page
When they click the "Tech Lead" persona button in the sidebar
Then all metric cards update to show Tech Lead data (18 features, 8 KPIs)
And the persona card displays "Tech Lead / Engineer"
And the Feature Coverage section shows 5 categories

NAV-007: Direct URL access to Dashboard

Given the user navigates directly to /dtq/dashboard
Then the Dashboard page loads with the default persona (Manager)
And all components render without errors

NAV-008: Direct URL access to Reports

Given the user navigates directly to /dtq/reports
Then the Reports page loads with test run data
And the table displays up to 10 rows per page

NAV-009: Direct URL access to History

Given the user navigates directly to /dtq/history
Then the History page loads with 30-day metric data
And all 4 trend charts render

11.2 Dashboard Tests

DASH-001: Primary metric card click opens MetricDrillDownModal

Given the user is on the Dashboard page
When they click the "Total Features" metric card
Then the MetricDrillDownModal opens
And it displays the label "Total Features"
And it shows a 30-day trend chart
And it includes period comparison data

DASH-002: Automation Rate metric card click

Given the user is on the Dashboard page
When they click the "Automation Rate" metric card
Then the MetricDrillDownModal opens
And it displays the automation rate value with "%" unit
And the trend shows "up"

DASH-003: Risk Distribution metric card click

Given the user is on the Dashboard page
When they click the "Risk Distribution" metric card
Then the MetricDrillDownModal opens
And it displays the high-risk count with "high risk" unit

DASH-004: Open Defects metric card click

Given the user is on the Dashboard page
When they click the "Open Defects" metric card
Then the MetricDrillDownModal opens
And it displays the open defects count

DASH-005: Persona KPI card click opens MetricDrillDownModal

Given the user is on the Dashboard with QA Manager persona
When they click the "Test Pass Rate" KPI card
Then the MetricDrillDownModal opens
And it displays "Test Pass Rate" with value 94.2%
And trend shows "up" with "+2.3%"

DASH-006: Pass Rate chart data point click

Given the user is on the Dashboard page
When they click a data point on the "Test Pass Rate Trend" chart
Then the ChartDrillDownModal opens
And it shows the value for the clicked date
And it displays comparison to previous/next days

DASH-007: Automation Coverage chart data point click

Given the user is on the Dashboard page
When they click a data point on the "Automation Coverage" chart
Then the ChartDrillDownModal opens
And it shows the automation coverage value for that date

DASH-008: Feature click opens FeatureDetailModal

Given the user is on the Dashboard with a category expanded
When they click a feature row (e.g., "Mobile Experience")
Then the FeatureDetailModal opens
And it shows the feature name, category, and status
And it displays 4 KPI cards (coverage, risk, defects, pass rate)
And it shows a 14-day history chart
And it lists recent test runs

DASH-009: Category click opens CategoryAnalyticsModal

Given the user is on the Dashboard page
When they click a category header (e.g., "Core User Journeys")
Then the CategoryAnalyticsModal opens

And it shows summary cards for the category
And it displays a feature comparison bar chart
And it shows a risk distribution pie chart

DASH-010: High Risk Banner feature click

Given the user is on the Dashboard with high-risk features present
When they click a feature in the High Risk Banner
Then the FeatureDetailModal opens for that feature
And the risk score is displayed (≥ 40)

DASH-011: Live toggle pause

Given the simulation is running (live)
When the user clicks the Live toggle
Then the indicator changes to paused state (grey dot)
And no new test runs are generated
And metrics stop fluctuating

DASH-012: Live toggle resume

Given the simulation is paused
When the user clicks the Live toggle
Then the indicator changes to live state (green pulsing dot)
And new test runs resume generating every 8–15 seconds
And metrics resume fluctuating

DASH-013: Feature Coverage search

Given the user is on the Dashboard page
When they type "mobile" in the Feature Coverage search input
Then only features matching "mobile" are displayed
And non-matching features are hidden

DASH-014: Cross-modal from CategoryAnalyticsModal to FeatureDetailModal

Given the CategoryAnalyticsModal is open
When the user clicks a feature in the category's feature list
Then the CategoryAnalyticsModal closes
And the FeatureDetailModal opens for the clicked feature

11.3 Reports Tests

RPT-001: Status filter — All tab

Given the user is on the Reports page
When they click the "All" filter tab
Then all test runs are displayed in the table
And the tab shows the total count

RPT-002: Status filter — Passed tab

Given the user is on the Reports page
When they click the "Passed" filter tab
Then only passed test runs are displayed
And every row shows a green check icon
And the tab count matches the visible rows

RPT-003: Status filter — Failed tab

Given the user is on the Reports page
When they click the "Failed" filter tab
Then only failed test runs are displayed
And every row shows a red X icon
And the issue column shows non-zero counts

RPT-004: Search by feature name

Given the user is on the Reports page
When they type "SCORM" in the search input
Then only test runs with "SCORM" in the feature name are shown
And the pagination updates to reflect the filtered count

RPT-005: Column sorting — ascending

Given the user is on the Reports page with no active sort
When they click the "Feature Name" column header
Then rows are sorted alphabetically A-Z by feature name
And an up arrow (↑) appears next to the column header

RPT-006: Column sorting — descending

Given the "Feature Name" column is sorted ascending
When the user clicks the "Feature Name" column header again
Then rows are sorted alphabetically Z-A
And a down arrow (↓) appears next to the column header

RPT-007: Column sorting — clear

Given the "Feature Name" column is sorted descending
When the user clicks the "Feature Name" column header a third time
Then the sort is cleared
And rows return to default order (most recent first)
And no sort arrow is displayed

RPT-008: Duration column sorting

Given the user is on the Reports page
When they click the "Duration" column header

Then rows are sorted by duration ascending (shortest first)

RPT-009: Executed column sorting

Given the user is on the Reports page
When they click the "Executed" column header
Then rows are sorted by execution date ascending (oldest first)

RPT-010: Issues column sorting

Given the user is on the Reports page
When they click the "Issues" column header
Then rows are sorted by issue count ascending (0 first)

RPT-011: Pagination — next page

Given the Reports table has more than 10 rows
When the user clicks the Next page button (▶)
Then the table shows rows 11–20
And the row indicator updates to "11–20 of N runs"

RPT-012: Pagination — previous page

Given the user is on page 2 of the Reports table
When they click the Previous page button (◀)
Then the table shows rows 1–10

RPT-013: Pagination — first page

Given the user is on page 3 of the Reports table
When they click the First page button (◀◀)
Then the table shows rows 1–10
And page 1 is highlighted in the page numbers

RPT-014: Pagination — last page

Given the user is on page 1 of the Reports table
When they click the Last page button (▶▶)
Then the table shows the final page of results

RPT-015: Pagination — page number click

Given the Reports table has multiple pages
When the user clicks page number "2"
Then the table shows rows 11–20
And page 2 is highlighted

RPT-016: Pagination — smart ellipsis

Given the Reports table has 5+ pages
And the user is on page 3
Then the page numbers show: 1 ... 2 3 4 ... 5

RPT-017: Row click opens TestRunDetailModal

Given the user is on the Reports page
When they click a test run row
Then the TestRunDetailModal opens
And it shows the feature name with status and date
And it displays test counts (total, passed, failed)

RPT-018: Cross-modal — test run to feature detail

Given the TestRunDetailModal is open
When the user clicks "View Feature"
Then the TestRunDetailModal closes
And the FeatureDetailModal opens for the associated feature

RPT-019: CSV export download

Given the user is on the Reports page
When they click "Export CSV"
Then a file named "test-reports-YYYY-MM-DD.csv" downloads
And it contains headers: ID, Feature, Status, Total Tests, Passed, Failed, Duration (s), Executed At
And every test run is included as a row

RPT-020: PDF export dialog

Given the user is on the Reports page
When they click "Export PDF"
Then the browser print dialog opens
And the print preview shows a styled test run table

RPT-021: Row highlight on newest run

Given the simulation is live
When a new test run is generated
Then the newest row receives a highlight animation
And the highlight fades after 1.5 seconds

RPT-022: Filter resets pagination

Given the user is on page 3 of the Reports table
When they click the "Failed" filter tab
Then the page resets to page 1
And only failed runs are shown

RPT-023: Search resets pagination

Given the user is on page 2 of the Reports table
When they type a search query
Then the page resets to page 1

RPT-024: Empty state display

Given the user is on the Reports page
When they search for a non-existent feature name (e.g., "xyznonexistent")
Then the table shows an empty state message
And the message says "No test runs match your filters"

11.4 History Tests

HIST-001: Summary card click — Average Pass Rate

Given the user is on the History page
When they click the "Average Pass Rate" summary card
Then the MetricDrillDownModal opens
And it displays "Average Pass Rate" with the 30-day average
And it shows a 30-day trend chart

HIST-002: Summary card click — First Pass Rate

Given the user is on the History page
When they click the "First Pass Rate" summary card
Then the MetricDrillDownModal opens
And it displays the first-run pass rate average

HIST-003: Summary card click — Defect Detection

Given the user is on the History page
When they click the "Defect Detection" summary card
Then the MetricDrillDownModal opens
And it displays the defect detection average

HIST-004: Summary card click — Test Effectiveness

Given the user is on the History page
When they click the "Test Effectiveness" summary card
Then the MetricDrillDownModal opens
And it displays the effectiveness average

HIST-005: Chart point click — Pass Rate Trend

Given the user is on the History page
When they click a data point on the "Test Pass Rate Trend" chart
Then the ChartDrillDownModal opens

And it shows the pass rate value for the clicked date
And it displays the percentile ranking

HIST-006: Chart point click — First Run Pass Rate

Given the user is on the History page
When they click a data point on the "First Run Pass Rate" chart
Then the ChartDrillDownModal opens
And the metric label says "First Run Pass Rate"

HIST-007: Chart point click — Defect Detection

Given the user is on the History page
When they click a data point on the "Defect Detection Percentage" chart
Then the ChartDrillDownModal opens
And the metric label says "Defect Detection"

HIST-008: Chart point click — Test Effectiveness

Given the user is on the History page
When they click a data point on the "Test Case Effectiveness" chart
Then the ChartDrillDownModal opens
And the metric label says "Test Effectiveness"

HIST-009: 30-day data validation

Given the user is on the History page
Then each of the 4 trend charts displays exactly 30 data points
And dates range from 2026-01-05 to 2026-02-03
And all values are within expected ranges per persona

11.5 Chat Tests

CHAT-001: Open chat widget

Given the chat widget is closed (FAB visible)
When the user clicks the floating action button
Then the chat panel slides open
And the welcome state is displayed ("Ask me about your testing metrics")
And the input field is auto-focused

CHAT-002: Close chat widget

Given the chat widget is open
When the user clicks the X close button
Then the chat panel closes
And the floating action button reappears

CHAT-003: Quick action pill — High-risk features

Given the chat widget is open
When the user clicks the "High-risk features" quick action pill
Then a user message "High-risk features" appears in the chat
And a typing indicator is shown
And an assistant response appears with high-risk feature analysis

CHAT-004: Quick action pill — Feature status

Given the chat widget is open
When the user clicks the "Feature status" quick action pill
Then a user message "Feature status" appears
And an assistant response appears with feature coverage summary

CHAT-005: Quick action pill — Automation gaps

Given the chat widget is open
When the user clicks the "Automation gaps" quick action pill
Then a user message "Automation gaps" appears
And an assistant response appears with automation coverage opportunities

CHAT-006: Quick action pill — Quality summary

Given the chat widget is open
When the user clicks the "Quality summary" quick action pill
Then a user message "Quality summary" appears
And an assistant response appears with executive quality summary

CHAT-007: Custom message send via Enter key

Given the chat widget is open
And the user has typed "What is the flaky test rate?" in the input
When the user presses Enter
Then the input clears
And a user message appears with the typed text
And a typing indicator shows while awaiting response
And an assistant response appears

CHAT-008: Custom message send via Send button

Given the chat widget is open
And the user has typed a question in the input
When the user clicks the Send button
Then the message is sent and the input clears

CHAT-009: Send button disabled when input empty

Given the chat widget is open
And the input field is empty

Then the Send button is visually disabled (50% opacity)
And clicking it does nothing

CHAT-010: Related link card click — navigate to feature

Given an assistant response includes a related link card for a feature
When the user clicks the link card
Then the app navigates to /dtq/dashboard
And the FeatureDetailModal opens for the linked feature

CHAT-011: Related link card click — navigate to category

Given an assistant response includes a related link card for a category
When the user clicks the link card
Then the app navigates to /dtq/dashboard
And the CategoryAnalyticsModal opens for the linked category

CHAT-012: Related link card click — navigate to report

Given an assistant response includes a related link card for a test run
When the user clicks the link card
Then the app navigates to /dtq/reports
And the TestRunDetailModal opens for the linked test run

CHAT-013: Related link card click — navigate to history metric

Given an assistant response includes a related link card for a history metric
When the user clicks the link card
Then the app navigates to /dtq/history
And the MetricDrillDownModal opens for the linked metric

CHAT-014: Cross-page chat persistence

Given the user has sent messages in the chat on the Dashboard page
When the user navigates to the Reports page via sidebar
Then the chat widget retains all previous messages
And the conversation history is preserved

CHAT-015: Reset button clears history

Given the chat widget has messages
When the user clicks the Reset button (rotating arrow icon)
Then all messages are cleared
And the welcome state reappears

CHAT-016: Reset button hidden when no messages

Given the chat widget is open with no messages
Then the Reset button is not visible

CHAT-017: Unread badge counter

Given the chat widget is closed
When a quick action or background response generates a new message
Then an unread badge appears on the floating action button
And the badge shows the unread count

CHAT-018: Unread badge overflow

Given the unread count exceeds 9
Then the badge displays "9+" instead of the actual number

CHAT-019: Source display

Given an assistant response was generated using RAG
Then below the response, a source indicator shows (e.g., "3 sources used")
And a paperclip icon is displayed

CHAT-020: Persona badge in header

Given the chat widget is open
Then the header shows a persona badge with the current persona name
And the badge icon matches the persona (Crown/Users/Code)

11.6 Cross-Page Navigation Tests

XNAV-001: Chat link → Dashboard feature modal

Given the user is on the Reports page
And the chat has a related link card targeting a feature on the Dashboard
When the user clicks the link card
Then the app navigates to /dtq/dashboard
And the FeatureDetailModal opens for the specified feature

XNAV-002: Chat link → Dashboard category modal

Given the user is on the History page
And the chat has a related link card targeting a category
When the user clicks the link card
Then the app navigates to /dtq/dashboard
And the CategoryAnalyticsModal opens for the specified category

XNAV-003: Chat link → Reports page test run

Given the user is on the Dashboard page
And the chat has a related link card targeting a test run
When the user clicks the link card
Then the app navigates to /dtq/reports
And the TestRunDetailModal opens for the matching test run

XNAV-004: Chat link → History page metric

Given the user is on the Dashboard page
And the chat has a related link card targeting a history metric (e.g., "passRate")
When the user clicks the link card
Then the app navigates to /dtq/history
And the MetricDrillDownModal opens for the specified metric

XNAV-005: Navigation action TTL expiration

Given a navigation action has been dispatched
When 30 seconds pass without the target page processing it (ACTION_TTL_MS = 30000)
Then the pending action is considered stale
And the target page does not open a modal for expired actions

11.7 Modal Tests

MOD-001: Modal close via X button

Given any modal is open
When the user clicks the X close button
Then the modal closes with an exit animation
And the underlying page is interactive again

MOD-002: Modal close via Escape key

Given any modal is open
When the user presses the Escape key
Then the modal closes

MOD-003: Modal focus trap

Given a modal is open
When the user presses Tab repeatedly
Then focus cycles through focusable elements within the modal
And focus does not escape to the background page

MOD-004: TestRunDetailModal severity filtering

Given the TestRunDetailModal is open for a failed test run
When the user clicks the "High" severity filter tab
Then only high-severity issues are displayed
And medium and low severity issues are hidden

MOD-005: TestRunDetailModal stack trace expand

Given the TestRunDetailModal is open with issues
When the user clicks on an issue's error message

Then the stack trace expands below the error message
And clicking again collapses the stack trace

MOD-006: TestRunDetailModal copy button

Given the TestRunDetailModal shows an issue with a stack trace
When the user clicks the copy button
Then the stack trace is copied to the clipboard

12. Appendix

12.1 All 80 Features

#	ID	Feature	Category	Persona	Coverage	Risk	Pass Rate
1	cs-f1	Payment Gateway	Revenue Platform	C-Suite	96%	12	98%
2	cs-f2	Subscription Billing	Revenue Platform	C-Suite	93%	18	96%
3	cs-f3	Revenue Analytics	Revenue Platform	C-Suite	91%	15	97%
4	cs-f4	Pricing Engine	Revenue Platform	C-Suite	88%	22	94%
5	cs-f5	Customer Portal	Customer Experience	C-Suite	94%	14	97%
6	cs-f6	NPS Tracking	Customer Experience	C-Suite	89%	20	93%
7	cs-f7	Support Escalation	Customer Experience	C-Suite	92%	16	96%
8	cs-f8	Onboarding Journey	Customer Experience	C-Suite	90%	19	95%
9	cs-f9	Multi-Region Deployment	Market Expansion	C-Suite	87%	24	92%
10	cs-f10	Localization Engine	Market Expansion	C-Suite	85%	28	90%
11	cs-f11	Partner Integration	Market Expansion	C-Suite	91%	17	95%
12	cs-f12	SOC 2 Validation	Strategic Compliance	C-Suite	97%	10	99%
13	cs-f13	GDPR Data Handling	Strategic Compliance	C-Suite	95%	13	98%

14	cs-f14	Audit Trail System	Strategic Compliance	C-Suite	93%	16	96%
15	cs-f15	AI Feature Pipeline	Digital Transformation	C-Suite	82%	30	89%
16	cs-f16	Legacy Migration	Digital Transformation	C-Suite	84%	26	91%
17	f1	xAPI / LRS Integration	Advanced / Enterprise	Manager	92%	45	94%
18	f2	Multi-Tenant Configuration	Advanced / Enterprise	Manager	88%	32	91%
19	f3	Advanced Compliance Tracking	Advanced / Enterprise	Manager	95%	18	97%
20	f4	Custom Branding Engine	Advanced / Enterprise	Manager	78%	25	89%
21	f5	Enterprise SSO	Advanced / Enterprise	Manager	96%	12	98%
22	f6	Adaptive Learning Paths	Learning Experience	Manager	85%	38	88%
23	f7	Personalized Recommendations	Learning Experience	Manager	91%	22	93%
24	f8	Skill Gap Analysis	Learning Experience	Manager	87%	28	90%
25	f9	Learning Goals Tracking	Learning Experience	Manager	93%	15	96%
26	f10	Executive Dashboard	Reporting & Analytics	Manager	94%	20	95%
27	f11	Custom Report Builder	Reporting & Analytics	Manager	82%	35	87%
28	f12	Learning Analytics	Reporting & Analytics	Manager	89%	42	91%
29	f13	Compliance Reports	Reporting & Analytics	Manager	97%	10	99%
30	f14	Export Center	Reporting & Analytics	Manager	90%	16	94%
31	f15	Course Discovery	Course Catalog & Enrollment	Manager	86%	30	89%

32	f16	Self-Enrollment	Course Catalog & Enrollment	Manager	95%	14	97%
33	f17	Manager-Assigned Learning	Course Catalog & Enrollment	Manager	88%	24	92%
34	f18	Waitlist Management	Course Catalog & Enrollment	Manager	79%	20	86%
35	f19	Discussion Forums	Social & Collaboration	Manager	83%	26	88%
36	f20	Peer Reviews	Social & Collaboration	Manager	91%	18	93%
37	f21	Learning Groups	Social & Collaboration	Manager	87%	22	90%
38	f22	Social Sharing	Social & Collaboration	Manager	76%	15	85%
39	f23	SCORM Import	Course Creation & Management	Manager	94%	19	95%
40	f24	Content Authoring	Course Creation & Management	Manager	81%	34	86%
41	f25	Assessment Builder	Course Creation & Management	Manager	89%	28	91%
42	f26	Media Library	Course Creation & Management	Manager	85%	21	89%
43	f27	Course Versioning	Course Creation & Management	Manager	92%	16	94%
44	f28	User Management	Admin & Configuration	Manager	96%	12	98%
45	f29	Role-Based Access Control	Admin & Configuration	Manager	93%	18	96%
46	f30	Notification Settings	Admin & Configuration	Manager	84%	22	88%
47	f31	Integration Hub	Admin & Configuration	Manager	78%	36	85%
48	f32	Audit Logs	Admin & Configuration	Manager	97%	10	99%
49	f33	System Configuration	Admin & Configuration	Manager	88%	24	91%
50	f34	Quiz Engine	Assessments & Certification	Manager	92%	20	94%

51	f35	Certification Paths	Assessments & Certification	Manager	90%	16	93%
52	f36	Proctoring Support	Assessments & Certification	Manager	75%	32	84%
53	f37	Certificate Designer	Assessments & Certification	Manager	86%	18	90%
54	f38	Webinar Integration	Events & Live Sessions	Manager	82%	28	87%
55	f39	ILT Scheduling	Events & Live Sessions	Manager	89%	22	92%
56	f40	Attendance Tracking	Events & Live Sessions	Manager	94%	14	96%
57	f41	Virtual Classroom	Events & Live Sessions	Manager	77%	38	85%
58	f42	User Onboarding	Core User Journeys	Manager	91%	24	93%
59	f43	Course Completion Flow	Core User Journeys	Manager	95%	12	97%
60	f44	Profile Management	Core User Journeys	Manager	88%	18	91%
61	f45	Mobile Experience	Core User Journeys	Manager	79%	42	86%
62	f46	Search & Discovery	Core User Journeys	Manager	84%	30	88%
63	tl-f1	Build Orchestration	CI/CD Pipeline	Tech Lead	88%	25	91%
64	tl-f2	Deploy Pipeline	CI/CD Pipeline	Tech Lead	85%	32	88%
65	tl-f3	Rollback Automation	CI/CD Pipeline	Tech Lead	82%	35	86%
66	tl-f4	Feature Flags	CI/CD Pipeline	Tech Lead	90%	20	93%
67	tl-f5	REST API Gateway	API Infrastructure	Tech Lead	92%	18	94%
68	tl-f6	GraphQL Layer	API Infrastructure	Tech Lead	79%	38	85%
69	tl-f7	Rate Limiting	API Infrastructure	Tech Lead	86%	22	92%

70	tl-f8	API Versioning	API Infrastructure	Tech Lead	83%	28	89%
71	tl-f9	Load Testing Framework	Performance Engineering	Tech Lead	78%	40	84%
72	tl-f10	Memory Profiler	Performance Engineering	Tech Lead	75%	36	82%
73	tl-f11	CDN Optimization	Performance Engineering	Tech Lead	91%	15	95%
74	tl-f12	Penetration Test Suite	Security Testing	Tech Lead	70%	45	80%
75	tl-f13	Dependency Scanner	Security Testing	Tech Lead	93%	16	94%
76	tl-f14	Auth Token Validation	Security Testing	Tech Lead	89%	20	92%
77	tl-f15	Container Orchestration	DevOps Automation	Tech Lead	84%	30	88%
78	tl-f16	Log Aggregation	DevOps Automation	Tech Lead	87%	19	91%
79	tl-f17	Infra-as-Code	DevOps Automation	Tech Lead	81%	33	86%
80	tl-f18	Secrets Management	DevOps Automation	Tech Lead	96%	10	98%

12.2 All 24 KPIs

#	Persona	Key	Label	Value	Unit	Trend
1	C-Suite	releaseVelocity	Release Velocity	+35	%	up
2	C-Suite	timeToMarket	Mean Time to Market	12	days	down
3	C-Suite	automationROI	Automation ROI	285	%	up
4	C-Suite	escapeRate	Defect Escape Rate	2.1	%	down
5	C-Suite	incidentsPrevented	Incidents Prevented	47	count	up
6	C-Suite	riskReduction	Risk Reduction	68	%	up
7	C-Suite	capacityUnlocked	Capacity Unlocked	42	%	up
8	Manager	passRate	Test Pass Rate	94.2	%	up
9	Manager	autoManualRatio	Automated vs Manual	8.5:1	ratio	up
10	Manager	effectiveness	Test Case Effectiveness	0.87	score	stable

11	Manager	regressionTime	Regression Execution	42	min	down
12	Manager	firstRunPass	First-Run Pass Rate	87.5	%	up
13	Manager	escalationRate	Escalation Rate	12.3	%	down
14	Manager	testReuse	Test Case Reuse	76.8	%	up
15	Manager	blockerDefects	Blocker Defects	3	count	down
16	Manager	envUptime	Environment Uptime	99.2	%	stable
17	Tech Lead	flakyRate	Flaky Test Rate	3.2	%	down
18	Tech Lead	avgExecution	Avg Execution Time	4.2	min	down
19	Tech Lead	tokenCost	Token Usage Cost	0.42	\$	down
20	Tech Lead	contextHitRate	Context Hit Rate	94.5	%	up
21	Tech Lead	toolCallSuccess	Tool Call Success	97.8	%	up
22	Tech Lead	parallelEfficiency	Parallel Efficiency	78.5	%	up
23	Tech Lead	automationCoverage	Automation Coverage	93.3	%	up
24	Tech Lead	totalTests	Total Test Cases	1247	count	up

12.3 All 20 Categories

#	Category	Persona	Feature Count
1	Revenue Platform	C-Suite	4
2	Customer Experience	C-Suite	4
3	Market Expansion	C-Suite	3
4	Strategic Compliance	C-Suite	3
5	Digital Transformation	C-Suite	2
6	Advanced / Enterprise	Manager	5
7	Learning Experience	Manager	4
8	Reporting & Analytics	Manager	5
9	Course Catalog & Enrollment	Manager	4
10	Social & Collaboration	Manager	4
11	Course Creation & Management	Manager	5
12	Admin & Configuration	Manager	6
13	Assessments & Certification	Manager	4

14	Events & Live Sessions	Manager	4
15	Core User Journeys	Manager	5
16	CI/CD Pipeline	Tech Lead	4
17	API Infrastructure	Tech Lead	4
18	Performance Engineering	Tech Lead	3
19	Security Testing	Tech Lead	3
20	DevOps Automation	Tech Lead	4

12.4 API Endpoint Reference

Endpoint	Method	Description
/api/dtq/features	GET	List all features for current persona
/api/dtq/categories	GET	Categories with aggregated stats
/api/dtq/test-runs	GET	Test run history with issues
/api/dtq/test-runs	POST	Create new test run (simulation)
/api/dtq/metrics	GET	Daily metrics (30 days)
/api/dtq/metrics?type=summary	GET	Summary metrics computed from features
/api/dtq/personas	GET	All personas with their KPI metrics
/api/dtq/chat	POST	Claude-powered AI chat with RAG pipeline

12.5 Knowledge Base Statistics

Item Type	Count	Description
feature	80	All features across 3 personas (16 + 46 + 18)
category_summary	20	Category aggregations (5 + 10 + 5)
persona_kpi	24	KPI metrics (7 + 9 + 8)
test_run_summary	3	Per-persona test execution summaries
daily_metrics_summary	3	Per-persona 30-day metrics overviews
Total	130	100% embedding coverage (1536-dim)

12.6 Database Schema

Table	Description	Row Count
dtq.features	Test features across categories	46 (Manager base)
dtq.test_runs	Test execution history	40+

dtq.test_issues	Failures/defects from test runs	30
dtq.daily_metrics	30-day historical metrics	30
dtq.personas	Role-based persona definitions	3
dtq.persona_metrics	KPIs per persona	24
dtq.knowledge_base	AI knowledge base with vector embeddings	130

12.7 Semantic Search RPC

Function: `public.search_dtq_knowledge_semantic`

Input: JSONB containing:

- `query_embedding` — 1536-dimensional vector from OpenAI
- `match_threshold` — Minimum cosine similarity (default: 0.5)
- `match_count` — Maximum results to return

Output: Matching knowledge base rows with similarity scores, ordered by relevance.

Part of Digital Workplace AI Product Suite Location: `/Users/aldrin-mac-mini/digitalworkplace.ai/apps/test-iq` Port: 3004 | BasePath: `/dtq` | Status: `IMPLEMENTED`